

Shram Kadia

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SUMMARY

Software engineer with deep experience in ML systems ? production data pipelines ? and observability infrastructure. Skilled in building retrieval-augmented generation systems ? semantic search engines ? and distributed tracing systems.

CORE COMPETENCIES

- ML Systems
- Distributed Tracing
- FastAPI
- HuggingFace
- Observability

WORK EXPERIENCE

OpenPRA Org Aug 2025 ? Present

Software Engineer

- Owned end-to-end delivery of a probabilistic fault-tree generator for nuclear safety analysis ? shipped from concept to production in 8 weeks using Python ? TypeScript ? Node.js ? and Docker with full CI/CD automation.
- Authored 25+ Jest unit tests and CI checks guaranteeing numerical stability and reproducible outputs ? maintaining full coverage on safety-critical probability calculations.
- Architected type-safe probability classes across packages ? enabling rapid testing and iteration loops.
- Redesigned XML processing pipeline with lazy-parse techniques ? cutting debugging cycles by 40% and accelerating risk-assessment turnaround.

NC State University Jul 2024 ? Mar 2025

Backend Engineer ? Research Assistant

- Designed and deployed a production RAG stack using LangChain ? FastAPI ? FAISS ? HuggingFace transformers ? and Redis ? achieving 85% extraction accuracy and sub-200ms latency under concurrent load.
- Engineered retrieval pipeline with semantic chunking ? confidence thresholds, and exponential-backoff retry logic ? eliminating 60% of manual work and saving the department 30 hours/week.
- Integrated distributed tracing and structured logging into the FastAPI service; used telemetry to identify bottlenecks and drive a 30% throughput improvement.

Resilient Tech Jan 2023 ? May 2023

Software Engineer

- Migrated critical ERP systems to RESTful Python APIs across 5 enterprise rollouts ? cutting production downtime by 30% and sustaining 99% uptime.
- Resolved 100+ production support tickets by tracing root causes through front-end and back-end workflows ? lifting CSAT by 12%.
- Built and maintained client-facing React interfaces alongside Python backend APIs ? delivering full-stack features using Git-based version control and Agile workflows.

PROJECTS

Travel Agent ATLAS | Python ? FastAPI ? LangChain ? OpenAI-Compatible Proxy

- Built runtime LLM switching across 4 models with unified OpenAI-compatible proxy.
- Tracked per-request token usage and cost ? enabling live cost/performance tradeoffs.
- Fired parallel Exa neural search queries ? eliminating retrieval latency from critical path.

ClinicFlow | Python ? FastAPI ? LangChain ? MCP ? Docker

- Built speech-to-SOAP pipeline converting voice to structured clinical notes.
- Implemented fault-tolerant output with JSON-only structured format and graceful fallback.
- Created stdio JSON-RPC 2.0 MCP server with 3 tools and 5 protocol-layer unit tests.

EDUCATION

M.S. Computer Science, North Carolina State University May 2025

GPA: 3.63/4

B.Tech Computer Engineering, Pandit Deendayal Energy University May 2023

GPA: 9.62/10

TECHNICAL SKILLS

Languages: Python ? TypeScript ? JavaScript ? SQL ? Java ML Frameworks: TensorFlow ? PyTorch ? HuggingFace ? FAISS

Cloud & Data: GCP ? AWS ? PostgreSQL ? Redis ? Docker Observability: Distributed Tracing ? Structured Logging